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INTERNSHIP TASK REPORT

ANALYTICAL METHOD WALKTHROUGH

Proposed Polarimetric Determination of Sucrose Content: Analytical Procedure and Validation Protocol

Method procedure, system suitability and ICH Q2(R2)-aligned validation protocol

Document control	Details
Project	Pharmaceutical Quality Control Internship | Task 02
Analyte material	Sucrose (illustrative raw-material assay)
Analytical technique	Optical rotation using a digital Polarimeter
Document status	Training / proposed protocol outline; requires laboratory verification and approval before use This document presents a proposed analytical procedure and validation protocol for the polarimetric determination of sucrose. The procedure has not been experimentally validated and must undergo laboratory verification, scientific review and approval before routine use.
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1. Objective and scope

To outline a polarimetric procedure for estimating sucrose content in a prepared aqueous sample and to define a fit-for-purpose validation plan. Sucrose is optically active and rotates plane-polarized light; observed rotation is related to concentration, optical path length, wavelength and temperature.

Intended use: illustrative assay of a sucrose raw-material sample. This document is not a substitute for the current applicable pharmacopoeial monograph, the supplier's approved specification, or a site-approved SOP. Confirm the material grade and intended use before adopting any limits.

2. Principle

A polarimeter measures the angle through which a sample rotates plane-polarised light. Specific optical rotation is calculated from observed rotation, tube length and solution concentration:

$$[\alpha]_D^{20} = \frac{\alpha}{l \times c}$$

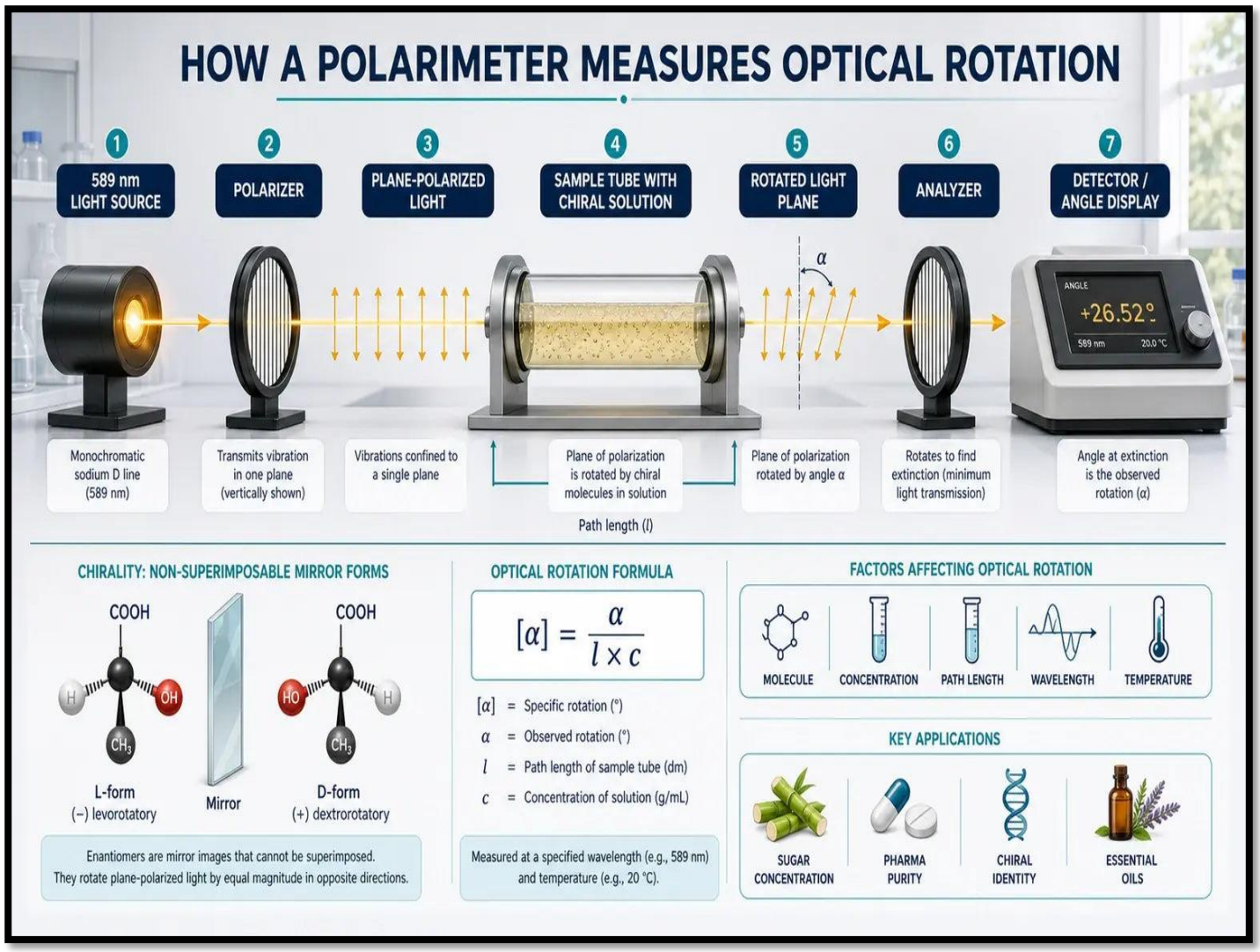
- $[\alpha]_D^{20}$: specific optical rotation at 20°C using sodium D-line light.
- α: observed optical rotation, in degrees.
- l: polarimeter tube length, in decimetres (dm).
- c: sucrose concentration, in g/mL.

Where α is observed rotation in degrees,

l is Polarimeter tube length in decimeters (dm), and

c is concentration in gm/mL.

The notation denotes the sodium D line (approximately 589 nm) at 20 °C. Any alternative wavelength or temperature must be stated and controlled.



https://scispectrum.in/blogs/guides/how-a-polarimeter-works-principles-of-optical-rotation?srsltid=AYCX9vLl_QeuUW5xQvvBT1dQXEGh_iehJP9Ttf6NmO5KeIKFbGU-g_-V

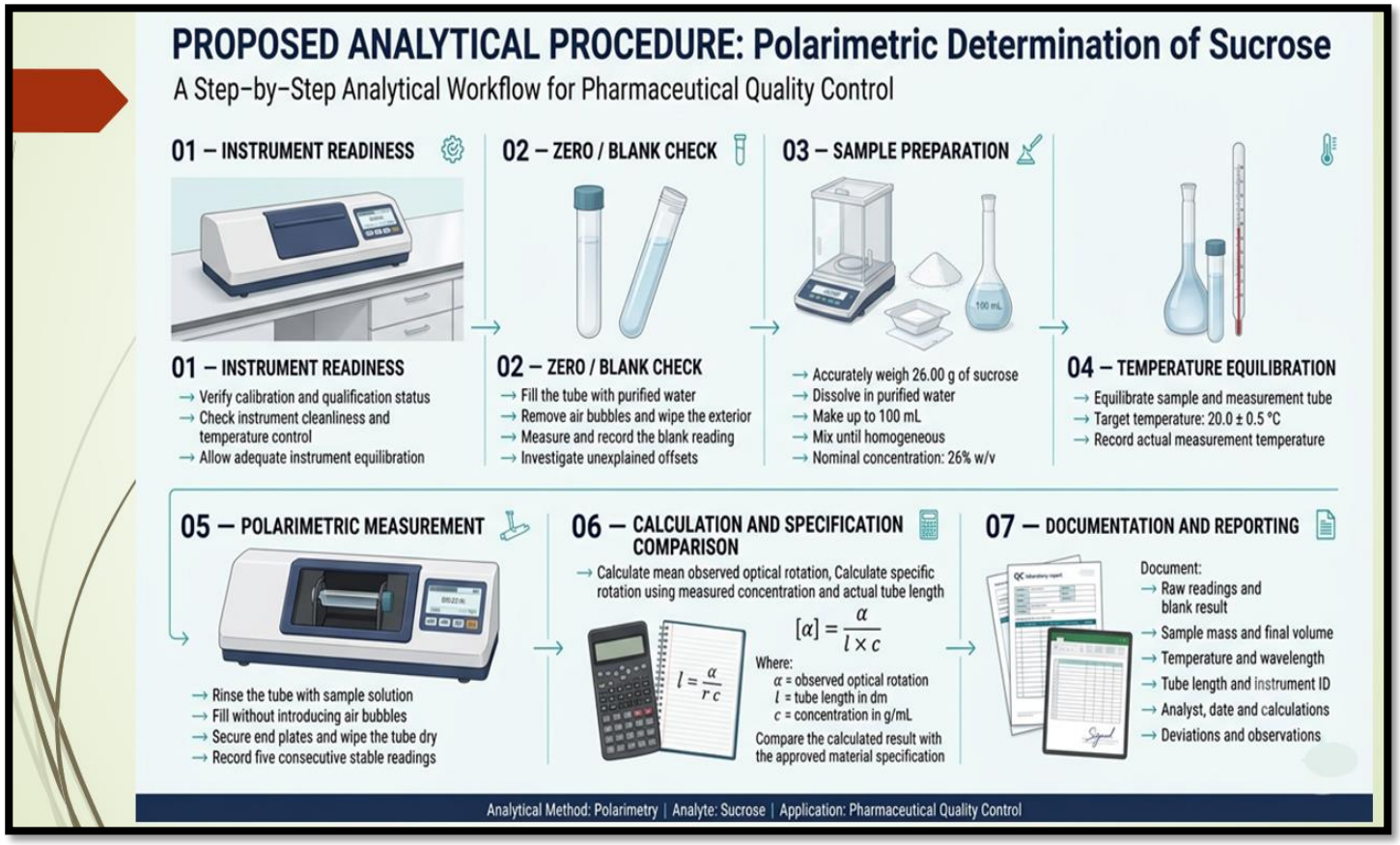
3. Equipment, materials and reagents

- Digital polarimeter capable of measurement at the sodium D line (approximately 589 nm), with temperature control or a verified temperature-controlled sample compartment.
- Matched, clean polarimeter tube of known path length (e.g., 2 dm); recorded the tube length.
- Analytical balance, Class A volumetric flask (100 mL), beaker, funnel and wash bottle.
- Purified water; certified or otherwise qualified sucrose reference material for validation and ongoing control checks.
- Test sample: sucrose raw material; record supplier, grade, batch/lot number and COA specification.

4. Proposed analytical procedure

Step	Procedure
1. Instrument readiness	Verifying calibration/qualification status, cleanliness, level and temperature control. Allowed the instrument to equilibrate as specified by its manufacturer.
2. Zero / blank check	Filling the clean tube with purified water, removing bubbles, wiping the exterior and measurement of the blank. Recording the reading. Re-zero only according to the instrument SOP; investigating unexplained offset.
3. Sample preparation	Accurately weighing 26.00 gm of sample into a 100 mL volumetric flask. Dissolution in purified water, dilution to volume at the controlled measurement temperature and mixing until homogeneous. Nominal concentration: 26% w/v.
4. Temperature equilibration	Bringing solution and tube to 20.0 ± 0.5 °C. Recording actual temperature by virtue of a consistent preparation-to-reading interval.
5. Measurement	Rinse the tube with a small portion of sample solution, fill without air bubbles, fit end plates, wipe dry and place in the polarimeter. Record five consecutive stable readings, or the number specified in the approved protocol.
6. Calculation	Calculate mean observed rotation and specific rotation using measured concentration and actual tube length. Compare with the applicable approved material specification.
7. Documentation	Record raw readings, blank, sample mass, final volume, temperature, wavelength, tube length, instrument ID, analyst, date, calculations and deviations.

Illustrative calculation only: For a 2 dm tube and c = 0.260 g/mL, if mean observed rotation is +34.58°, specific rotation = 34.58 / (2 × 0.260) = +66.50°. *This is not an experimental result*.



5. System suitability / routine performance checks

Check	Proposed criterion	Action if failed
Instrument zero / water blank	Blank within ±0.02° of zero, or tighter instrument-specific limit established during qualification.	Check tube cleanliness, bubbles, temperature and instrument zero; investigate before sample testing.
Reference standard check	Qualified sucrose reference gives specific rotation within its certified range / assigned control limits.	Do not report sample results; investigate standard, preparation, temperature and instrument.
Repeatability of readings	Five stable readings on one preparation: range ≤0.05° or RSD ≤0.5%, as appropriate to instrument capability.	Refill tube, remove bubbles, verify stability and repeat only after documented investigation.

Temperature	20.0 ± 0.5 °C throughout measurement.	Re-equilibrate and repeat measurement.
Visual inspection	Clear, homogeneous solution; no visible particles or air bubbles in tube.	Re-prepare or clarify using a justified procedure; document filtration because it may affect recovery.

These are proposed training criteria, not universal ICH limits. Final SST limits must be justified using instrument capability, reference-material data, method development and the applicable specification.

6. ICH Q2(R2)-aligned validation protocol outline

For an assay procedure, ICH Q2(R2) normally expects evaluation of specificity, response/calibration model and range, accuracy, repeatability and intermediate precision. Robustness is generally examined during development (with Q14 principles) and is included here as a supporting study. Detection and quantitation limits are generally not applicable to an assay whose reportable range is well above those limits.

Parameter	Proposed study design	Proposed acceptance criteria
Specificity / selectivity	Measure water blank, sucrose reference and sample. Challenge with plausible optically active interferents (e.g., glucose/fructose) where relevant; assess their contribution.	Blank has no material rotation; response is attributable to sucrose for the intended matrix. If other optically active substances may be present, polarimetry alone is insufficiently specific; use an orthogonal method.
Linearity / response	At least five levels spanning 80–120% of nominal (e.g., 20.8, 23.4, 26.0, 28.6 and 31.2 g/100 mL); measure in replicate and plot observed rotation against concentration.	Correlation coefficient $r \geq 0.999$; response is appropriately linear across range; residuals show no meaningful trend. Evaluate slope and intercept.
Range	Use interval supported by acceptable linearity, accuracy and precision; proposed range 80–120% of nominal concentration.	Accuracy and precision criteria are met throughout the proposed range.

Accuracy	Recovery study at 80%, 100% and 120% of nominal; three independent preparations per level (nine results).	Mean recovery 98.0–102.0% at each level; report individual recoveries, mean and confidence interval.
Repeatability	Prepare and analyse six independent sample preparations at 100% nominal under same conditions, analyst and instrument.	% RSD of assay results ≤1.0%.
Intermediate precision	Repeat six-preparation study on another day and, where feasible, with a second analyst and/or second qualified polarimeter.	Overall % RSD ≤ 2.0%; no practically meaningful analyst/day/instrument effect.
Robustness	Vary controlled factors: temperature (19.5/20.5 ℃), equilibration time and, where feasible, instrument/tube. Do not vary wavelength if using fixed sodium D source.	Assay remains within predefined material specification; mean result changes by ≤1.0% relative from nominal condition; SST passes.
Detection / quantitation limits	Not normally required for a high-level quantitative assay; document rationale. Reconsider if intended use changes to trace-level measurement.	Not applicable for this proposed assay purpose; do not claim DL/QL without a suitable study.

7. Validation execution and reporting sequence

Approved protocol before testing: objective, scope, responsibilities, equipment, reference standards, sample plan, calculations, statistical approach and predefined criteria.

Confirmed instrument qualification and reference-standard traceability; document method-development evidence and rationale for selected conditions.

Execution of specificity, response/linearity, range, accuracy, repeatability and intermediate-precision studies. Record raw data contemporaneously.

Evaluated robustness as a supporting development study. Investigate deviations and outliers using a predefined procedure; do not delete results without documented scientific justification.

Preparation of a validation report summarising data, calculations, deviations, statistical evaluation, criterion-by-criterion conclusions and limitations.

Approval for the final method and establish routine controls, analyst training, change control and periodic review before GMP use.

8. Important scientific limitation

Polarimetry measures net optical rotation from all optically active species in solution; it does not separate sucrose from other optically active sugars or chiral impurities. It is most defensible for a suitably controlled sucrose material with a known matrix. If other optically active compounds may be present, specificity must be supported by an orthogonal technique (for example, a suitable chromatographic assay), or the method must not be represented as sucrose-specific.

9. Acceptance-criteria summary

Proposed validation targets:

linearity $r \geq 0.999$;

accuracy 98.0–102.0%;

repeatability %RSD ≤1.0%;

intermediate precision overall % RSD ≤2.0%;

robustness change ≤1.0% relative under tested conditions;

and all defined SST checks pass before sample results are accepted. These are protocol proposals requiring scientific justification and confirmation against the applicable material specification.

10. References

1. International Council for Harmonization (ICH). ICH Q2(R2): Validation of Analytical Procedures. Final guideline, adopted 1 November 2023. https://database.ich.org/sites/default/files/ICH_Q2(R2)_Guideline_2023_1130.pdf
2. ICH Q14: Analytical Procedure Development https://www.fda.gov/media/161202/download
3. European Medicines Agency. ICH Q2(R2) Validation of analytical procedures – scientific guideline. https://www.ema.europa.eu/en/scientific-guidelines/ich-q2r2-validation-analytical-procedures
4. United States Pharmacopeia. General Chapter <781>, Optical Rotation. https://doi.org/10.31003/USPNF_M99570_03_01
5. SUCROSE - Specific Optical Rotation Standard traceable to NIST https://www.lobachemie.com/LaboratoryChemicals-S6157/sucrose--specific-optical-rotation-standard-traceable-to-nist-CASNO-57-50-1.aspx
6. Sucrose - British Pharmacopoeia https://nhathuocngocanh.com/bp/wp-content/uploads/2025/11/Sucrose-British-Pharmacopoeia.pdf
7. How a Polarimeter Works - Principles of Optical Rotation https://scispectrum.in/blogs/guides/how-a-polarimeter-works-principles-of-optical-rotation?srsltid=AYCX9vLl_QeuUW5xQvvBT1dQXEGh_iehJP9Ttf6NmO5KeIKFbGU-g_-V